

FORM TP 2010011



TEST CODE **01329010**

JANUARY 2010

**CARIBBEAN EXAMINATIONS COUNCIL
SECONDARY EDUCATION CERTIFICATE
EXAMINATION**

INFORMATION TECHNOLOGY

**Paper 01 – Technical Proficiency
THEORY**

$1\frac{1}{2}$ hours

18 JANUARY 2010 (a.m.)

INSTRUCTIONS TO CANDIDATES

- 1. This paper has THREE sections and a total of FOURTEEN questions. Candidates MUST answer ALL questions in all THREE sections.**
- 2. Number EACH answer correctly in the answer booklet.**
- 3. You may write Basic or Pascal (or any other language) code in those questions requiring pseudocode/algorithms.**

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SECTION I
HARDWARE AND SOFTWARE – 30 marks

Answer ALL questions.

1. Computers are made up of different components and the names of the components are often written as acronyms.

Write the meaning of EACH of the following computer acronyms and state ONE task that is performed by EACH of the components.

- (a) CU (2 marks)
- (b) ALU (2 marks)
- (c) RAM (2 marks)
- (d) ROM (2 marks)

Total 8 marks

2. State the type of optical disk that is suitable in EACH of the following situations:

- (a) Best for storing videos (1 mark)
- (b) Can be reused as necessary (1 mark)
- (c) Commonly used for distributing software (1 mark)

Total 3 marks

3. Figures A to C represent three different data capture methods.



Figure A



Figure B

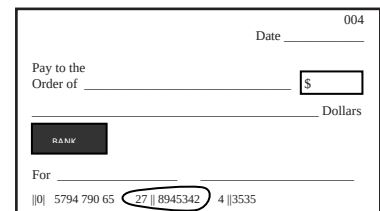


Figure C

In your answer booklet, copy and complete Table 1 using the list of type of data capture method below.

OMR, OCR, MICR, Voice Recognition

TABLE 1

Figure	Type of Data Capture Method
A	
B	
C	

Total 3 marks

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4. (a) You wish to purchase one of the devices shown in Figures 1 to 4.



Figure 1



Figure 2



Figure 3



Figure 4

Table 2 lists three requirements an organisation may need. In your answer booklet, copy and complete Table 2 to show which device would BEST meet EACH of the requirements in the table.

TABLE 2

Organisational Requirements	Figure
Useful for multi-form sheets	
Produces multiple copies rather quickly	
Ideal for coloured flyers	

(3 marks)

- (b) Write the general name given to the devices in part (a).

(1 mark)

Total 4 marks

5. Compute the following:

- (a) The eight-bit binary representation of the decimal number 17. **(1 mark)**

- (b) The sign-and-magnitude representation for the decimal number NEGATIVE 17. **(1 mark)**

- (c) The result of $55 - 25$, using eight-bit, two's complement representation and express your answer as an eight-bit binary number. **(1 mark)**

- (d) What is the decimal equivalent for the binary coded decimal number 0011 0101 1000? **(1 mark)**

Total 4 marks

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6. Answer the following questions that relate to user interfaces:

- (a) Write TWO ways in which a user can input a command using a graphical user interface.
(2 marks)
- (b) Name TWO types of menu interfaces.
(2 marks)
- (c) Explain why a graphical user interface is considered to be very user-friendly.
(2 marks)
- (d) Explain how a user can input a command using the DOS operating system.
(2 marks)

Total 8 marks

SECTION II

APPLICATIONS AND IMPLICATIONS – 30 marks

Answer ALL questions.

7. THREE friends have purchased computers to help them in the music industry.
- (a) State TWO uses of computers in the music industry. **(2 marks)**
 - (b) Name TWO devices that can connect to the computer and state what they are used for. **(4 marks)**
- Total 6 marks**
8. (a) State the differences between ‘.edu’, ‘.org’ and ‘.gov’ as seen in website addresses. **(3 marks)**
- (b) Write the name of a well-known search engine. **(1 mark)**
- (c) You wish to access <http://www.caribbeannews.com>.
- (i) State where you would type <http://www.caribbeannews.com> to access this site. **(1 mark)**
 - (ii) State what **http** and **www** represent in (c) above. **(2 marks)**
- Total 7 marks**
9. For EACH of the following, state whether it is (A), a private network, (B), a public network, or (C), private with selected access from public network users:
- (a) Intranet **(1 mark)**
 - (b) Extranet **(1 mark)**
 - (c) Internet **(1 mark)**
- Total 3 marks**
10. (a) Describe TWO ways in which a computer can be used illegally. **(4 marks)**
- (b) Outline ONE example of how an organization can misuse its customers’ personal data. **(2 marks)**

Total 6 marks

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- 11.** (a) Give ONE advantage and ONE disadvantage of using credit or debit cards to purchase items. **(2 marks)**
- (b) Explain TWO potential health problems that can develop from using a computer over prolonged periods. **(4 marks)**
- (c) Give ONE positive impact and ONE negative impact that is the result of increased use of cell phones in the workplace. **(2 marks)**

Total 8 marks

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SECTION III

PROGRAMMING – 30 marks

Answer ALL questions.

12. (a) Explain the difference between a 'syntax error' and a 'logic error'. **(2 marks)**
- (b) Explain the difference in the execution of the TWO lines of code written below:
- (i) Line 1: If MARK > 50 Then print "Good" **(1 mark)**
- (ii) Line 2: If MARK >= 50 Then print "Day" **(1 mark)**
- (c) The number 50 was the value for MARK.
- (i) State the line number corresponding to the above code that would print a result. **(1 mark)**
- (ii) Write the result that would be printed. **(1 mark)**

Total 6 marks

13. Answer the questions below using the following part of a program.

```
1    value1, value2: integer;
2    Minimum: real;
3
4    value1 = 30;
5    value2 = 25.9;
6
7    IF (value1 > value2)
8    THEN read('MinA is', value2)
9    ELSE writeln('MinB is', value2);
10
```

- (a) Write the line numbers that contain code to identify
- (i) input **(1 mark)**
- (ii) declaration **(2 marks)**
- (iii) output. **(1 mark)**
- (b) The program contains FOUR errors. For EACH error write the line number and the corrected line of code. **(4 marks)**

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- (c) Identify the following:
- (i) A variable (1 mark)
 - (ii) A conditional statement (1 mark)
 - (iii) A data type (1 mark)
- (d) Explain the purpose of the program segment. (3 marks)
- (e) State the result when EACH of the following pairs of numbers is used in the corrected program:
- (i) value1 = 40, value2 = 50 (2 marks)
 - (ii) value1 = 50, value2 = 40 (2 marks)
 - (iii) value1 = 50, value2 = 50 (2 marks)

Total 20 marks

- 14.** If **Box1** contains the number **16** and **Box2** contains the number **44**, state whether EACH of the following statements is TRUE or FALSE.

- (a) (Box1 = 16) (1 mark)
- (b) (Box1 < Box2) (1 mark)
- (c) (Box1 > 16) AND (Box1 < Box2) (1 mark)
- (d) NOT (Box2 = 44) (1 mark)

Total 4 marks

END OF TEST